IB Physics Exam Solutions - May 2026 Time Zone C (Americas)
Prepared by danny.doucette@gmail.com

HL Paper 1A

Question	Answer	Explanation
1	B	You could use $s = \frac{1}{2}(u + v)t$ with the initial velocity 0 Or draw a position-time graph and find the area under a triangle with height v and base t
2	A	In the vertical direction, the ball's velocity will be less because of air resistance, so the rock will reach the ground first. In the horizontal direction, the ball's velocity will be less because of air resistance, so the rock will have a greater range. To visualize this, imagine the ball is a balloon.
3	A	From a free-body diagram, we see that $F_B = F_D + F_g$. From the data booklet, $F_B = \rho V g$ but not that this density is the density of the fluid (i.e., water, not oil) and the volume is the volume of the object (i.e., the oil droplet). Since $\rho = m/V$, find $V = m/\rho_{oil}$ and when we put it together we get $\rho_{water} \frac{m}{\rho_{oil}} g = F_D + mg$. Rearranging to solve for the desired ratio, we get $\frac{\rho_{oil}}{\rho_{water}} = \frac{mg}{mg+F_D} = \frac{3}{3+2} = \frac{3}{5}$. Note that the ratio must be less than 1 in order for the bubble to rise.
4	B	Impulse is $J = F\,\Delta t = \Delta p$. Since we are given Δp that is the answer. The given 3.0 s is a red herring.
5	D	Using the work-energy theory, at constant velocity, $W = \Delta E_p$ and so we find $F_f\, s = F_g\, h$, which rearranges to $F_f = F_g\, h / s = 1.5\,kN \times 14\,m / 30\,m$.
6	D	The power is $P = W / t = F_{net}\, s / t = F_{net}\, v$. The net upward force is given by $2\,F\,\cos\theta$.
7	C	This is best done by elimination. If the clay is dropped at r = 0, there will be no change to the rotational kinetic energy. If the clay is dropped at r = R there will be a large change in the rotational kinetic energy. A full solution, beyond what would be expected on this exam, would be $\Delta E_{k,rot} = \frac{L^2}{2(\frac{1}{2}MR^2+mr^2)} - \frac{L^2}{2(\frac{1}{2}MR^2)} = \frac{-L^2 \times mr^2}{2(\frac{1}{2}MR^2+mr^2)\frac{1}{2}MR^2}$, which looks like the function $y = x^2/(1 + x^2)$ when plotted on a GDC.

8	C	Both torque and angular acceleration will be constant with time. The angular momentum is proportional to time because $L = I\omega$ and $\omega = \alpha t$. Rotational kinetic energy is proportional to time^2 because $E_{k,rot} = L^2/2I$.
9	D	Proper time is measured by a clock that is stationary in the rest frame of the spaceship that is at event E and F.
10	C	Use $P = \epsilon \sigma A T^4$ to find that $\frac{P_2}{P_1} = \frac{\epsilon \sigma (A/4) (2T)^4}{\sigma A T^4}$
11	A	You could set $k_1 A \frac{T_M - T_1}{\Delta x} = k_2 A \frac{T_2 - T_M}{\Delta x}$ and rearrange to find the answer. However, it is probably easier to imagine that $k_1 = k_2$: then, T_M should be the average of T_1 and T_2. Only answer A reduces to this result.
12	C	Statement II refers to the greenhouse effect, which causes Earth's surface to both absorb and radiate a higher intensity than it would with no atmosphere. Statement III refers to the result, which is that the greenhouse effect keeps Earth warmer.
13	C	From the idea gas law, the gradient $q = N k_B T$. But the internal energy is $U = \frac{3}{2} N k_B T$.
14	B	Remembering that adiabats on the PV diagram are steeper than isotherms is helpful here.
15	C	The equation is $V = W/q$. You need to remember that the side of the cell with the longer line is positive. Convention current flows clockwise, so a negative charge would travel counterclockwise.
16	B	The two resistors are in parallel, so the total resistance is $\frac{1}{2}R$. Thus, the power dissipated is $P = V^2 / R = \varepsilon^2 / \frac{1}{2}R$. The dissipated power should be higher with two resistors than with just one, since they are in parallel.
17	D	A straightforward application of $x = x_0 \sin(\omega t + \phi)$ and $a = -\omega^2 x$
18	B	This question relies on knowing that electromagnetic waves are doubly transverse: they are composed of electric and magnetic fields oscillating at right angles, and the direction of the wave's propagation is perpendicular to both of those.
19	D	The refraction angles are measured from the perpendicular, so we should use 30° and 40°. Then, it's $1.5 \sin 30° = n_2 \sin 40°$.

20	A	Slit width b causes the single slit diffraction pattern with a minimum at 0.05 rad. Distance between slits d causes the diffraction pattern with angle between bright fringes 0.01 rad. Since $b = \lambda / \theta$ and $d = n\lambda / \theta$ (in the small angle approximation, $\sin\theta \approx \theta$), $b/d = \frac{\lambda / 0.05}{1\lambda / 0.01} = 1/5$. Alternatively, once you know the ratio of angles is 1:5, A is the correct answer because the slit width must be smaller than the separation between the slits.
21	C	Draw the pipe, and note that the length of the pipe must be 15 cm. Then draw the next harmonic, which has one node at the closed end and one node ⅔ of the way along the pipe. It must have a wavelength of 20 cm.
22	B	The Doppler effect: light from objects moving away are redshifted, or have greater wavelengths.
23	A	We have a moving source so the frequency will be $f' = f\frac{v}{v - \frac{v}{5}} = \frac{5}{4}f$. Thus, the wavelength will be $\lambda' = c / f' = c / \frac{5}{4}f = \frac{4}{5}\lambda$.
24	A	The gravitational field strength equals the free fall acceleration, so $g = GM / r^2 = G\rho\frac{4}{3}\pi r^3 / r^2 = G\rho\frac{4}{3}\pi r$ so halving the radius will decrease g to half
25	C	First, find the distance to the object with mass $4m$. Since $g = GM/R^2$ is the same for the two masses, the $4m$ mass will be at a distance $2r$ from point Z. Then, the gravitational potential will be $V_P = \frac{-Gm}{r} + \frac{-G4m}{2r} = -\frac{3Gm}{r}$
26	B	Before they are in contact, the force between the spheres will be $F = \frac{k(Q)(-2Q)}{r^2} = -2\frac{kQ^2}{r^2}$. After they are in contact, each sphere will have charge $-\frac{1}{2}Q$ and so the force will be $F = \frac{k(-\frac{1}{2}Q)(-\frac{1}{2}Q)}{r^2} = +\frac{1}{4}\frac{kQ^2}{r^2}$. Thus, the new force is F/8 and the two spheres will both have negative charge and repel.
27	D	One way to think about it: between a positive and negative charge, there must be an equipotential with V = 0, which goes off to infinity. These are the vertical equipotential lines in the diagram. Only D has such an equipotential that divides the charges into a group of 3 and a group of 1.
28	A	In a uniform electric field, an electron will experience a uniform linear acceleration. This can manifest as a parabola or a straight line. There is a useful analogy to projectile motion here.

29	B	Wire Y will create a magnetic field to the left at the location of wire X. This is antiparallel to the current in wire X, so the magnetic force will be 0. However, a bit to the right of wire Y, the magnetic field from wire Y will have a component that is into the page. This will produce an upward force on the right-moving current in wire X. Likewise, a bit to the left of wire Y, the magnetic field from wire Y will have a component that is out of the page. This will produce a downward force on the right-moving current in wire X. The result is a counter-clockwise torque applied to wire X.
30	A	The negative electron moving right through a B-field out of the page will experience an upward magnetic force. Because the emf is constant, there will be an accumulation of electrons at the top of the rod, and so there will be a downward electric force.
31	D	emf is proportional to angular speed, $\varepsilon \propto \omega$. Since $\omega = 2\pi / T$, decreasing the period T will increase the emf. The other three changes will decrease the emf by decreasing B and A in $\Phi = BA\cos\theta$.
32	D	Most of the alpha particles pass straight through, so gold atoms must be largely empty
33	C	Downward transitions result in the emission of electromagnetic radiation. The larger transition results in the emission of a photon with more energy, i.e., the UV light.
34	A	In the formula $\Delta\lambda = \frac{h}{m_e c}(1 - \cos\theta)$, the angle is measured between the direction of the incident photon and the scattered photon.
35	B	The de Broglie wavelength is $\lambda = h / p$, and kinetic energy is $K = p^2 / 2m$. Thus, $\lambda = h / \sqrt{2\,m\,K}$.
36	D	The reaction equation will be $^{225}_{89}Ac \rightarrow {}^{A}_{Z}X + {}^{4}_{2}He$. Thus, the mass number A = 221 and the atomic number Z = 87. The number of protons is Z = 87, and the number of protons + neutrons is A = 221, so the neutrons is 221 - 87 = 134.
37	B	Beta plus decay involves the emission of an anti-electron, which must be accompanied by a neutrino (not an anti-neutrino). This one you just needed to remember.
38	C	The probability of decay per unit time is given by the decay constant λ. Since $T_{\frac{1}{2}} = \frac{ln\,2}{\lambda}$, the probability of decay in the next second will be $\frac{ln\,2}{28\,hours} = \frac{ln\,2}{28 \times 3600}$
39	A	A moderator should have a mass comparable to a neutron, because this will be most efficient at slowing down neutrons that hit it.

40	D	The parallax angle is $\phi/2$

SL Paper 1A

Question	Answer	Explanation
1	B	You could use $s = \frac{1}{2}(u + v)t$ with the initial velocity 0 Or draw a position-time graph and find the area under a triangle with height v and base t
2	A	In the vertical direction, the ball's velocity will be less because of air resistance, so the rock will reach the ground first. In the horizontal direction, the ball's velocity will be less because of air resistance, so the rock will have a greater range. To visualize this, imagine the ball is a balloon.
3	A	From a free-body diagram, we see that $F_B = F_D + F_g$. From the data booklet, $F_B = \rho V g$ but not that this density is the density of the fluid (i.e., water, not oil) and the volume is the volume of the object (i.e., the oil droplet). Since $\rho = m/V$, find $V = m/\rho_{oil}$ and when we put it together we get $\rho_{water} \frac{m}{\rho_{oil}} g = F_D + mg$. Rearranging to solve for the desired ratio, we get $\frac{\rho_{oil}}{\rho_{water}} = \frac{mg}{mg+F_D} = \frac{3}{3+2} = \frac{3}{5}$. Note that the ratio must be less than 1 in order for the bubble to rise.
4	B	Impulse is $J = F\,\Delta t = \Delta p$. Since we are given Δp that is the answer. The given 3.0 s is a red herring.
5	D	Using the work-energy theory, at constant velocity, $W = \Delta E_p$ and so we find $F_f\,s = F_g\,h$, which rearranges to $F_f = F_g\,h\,/\,s = 1.5\,kN \times 14\,m\,/\,30\,m$.
6	D	The power is $P = W\,/\,t = F_{net}\,s\,/\,t = F_{net}\,v$. The net upward force is given by $2\,F\,\cos\theta$.
7	C	Use $P = \epsilon\,\sigma\,A\,T^4$ to find that $\frac{P_2}{P_1} = \frac{\epsilon\,\sigma\,(A/4)\,(2T)^4}{\sigma\,A\,T^4}$
8	A	You could set $k_1\,A\,\frac{T_M - T_1}{\Delta x} = k_2\,A\,\frac{T_2 - T_M}{\Delta x}$ and rearrange to find the answer. However, it is probably easier to imagine that $k_1 = k_2$: then, T_M should be the average of T_1 and T_2. Only answer A reduces to this result.

9	C	Statement II refers to the greenhouse effect, which causes Earth's surface to both absorb and radiate a higher intensity than it would with no atmosphere. Statement III refers to the result, which is that the greenhouse effect keeps Earth warmer.
10	C	From the idea gas law, the gradient $q = N k_B T$. But the internal energy is $U = \frac{3}{2} N k_B T$.
11	C	The equation is $V = W/q$. You need to remember that the side of the cell with the longer line is positive. Convention current flows clockwise, so a negative charge would travel counterclockwise.
12	B	The two resistors are in parallel, so the total resistance is $\frac{1}{2}R$. Thus, the power dissipated is $P = V^2 / R = \varepsilon^2 / \frac{1}{2}R$. The dissipated power should be higher with two resistors than with just one, since they are in parallel.
13	B	This question relies on knowing that electromagnetic waves are doubly transverse: they are composed of electric and magnetic fields oscillating at right angles, and the direction of the wave's propagation is perpendicular to both of those.
14	D	The refraction angles are measured from the perpendicular, so we should use 30° and 40°. Then, it's $1.5 \sin 30° = n_2 \sin 40°$.
15	C	Draw the pipe, and note that the length of the pipe must be 15 cm. Then draw the next harmonic, which has one node at the closed end and one node ⅔ of the way along the pipe. It must have a wavelength of 20 cm.
16	B	The Doppler effect: light from objects moving away are redshifted, or have greater wavelengths.
17	A	The gravitational field strength equals the free fall acceleration, so $g = G M / r^2 = G \rho \frac{4}{3} \pi r^3 / r^2 = G \rho \frac{4}{3} \pi r$ so halving the radius will decrease g to half
18	B	Before they are in contact, the force between the spheres will be $F = \frac{k (Q) (-2Q)}{r^2} = - 2\frac{kQ^2}{r^2}$. After they are in contact, each sphere will have charge $-\frac{1}{2}Q$ and so the force will be $F = \frac{k (-\frac{1}{2}Q) (-\frac{1}{2}Q)}{r^2} = + \frac{1}{4}\frac{kQ^2}{r^2}$. Thus, the new force is F/8 and the two spheres will both have negative charge and repel.
19	A	In a uniform electric field, an electron will experience a uniform linear acceleration. This can manifest as a parabola or a straight line. There

		is a useful analogy to projectile motion here.
20	D	Use $\frac{F}{L} = \mu_0 \frac{I_1 I_2}{2\pi r} = 4\pi \times 10^{-7} \frac{3.0 \times 1.5}{2\pi\, 0.3} = 3.0 \times 10^{-6} N.$ You need to recall that parallel wires attract, or draw a diagram and use your hand rules.
21	D	Most of the alpha particles pass straight through, so gold atoms must be largely empty
22	C	Downward transitions result in the emission of electromagnetic radiation. The larger transition results in the emission of a photon with more energy, i.e., the UV light.
23	D	The reaction equation will be ${}^{225}_{89}Ac \rightarrow {}^{A}_{Z}X + {}^{4}_{2}He$. Thus, the mass number A = 221 and the atomic number Z = 87. The number of protons is Z = 87, and the number of protons + neutrons is A = 221, so the neutrons is 221 - 87 = 134.
24	A	A moderator should have a mass comparable to a neutron, because this will be most efficient at slowing down neutrons that hit it.
25	D	The parallax angle is $\phi/2$

HL and SL Paper 1B

1a	A straight line with negative gradient passes through all the error bars And there is a positive y-intercept
1b	$\frac{6.3 - 2.5}{0.04 - 0.24} = -19$
1c	$\frac{19}{9.81 \times \pi \times 0.028^2} = 786$
1d	The y-intercept is 7.1 N, and since this is the weight of the cylinder it will be equal to $\rho_w A\, L\, g$. Combine with the slope, which equals $\rho_l A\, g$, and we can find that $\rho_w / \rho_l = - y - int / L \times slope = 7.1 / 0.30 \times 19 = 1.25$
2ai	3.61 Ω
2aii	At (54, 3.61), and a straight line that passes through the error bars
bi	The rate of heat loss is decreasing over time
bii	Instead of making a measurement every 5 minutes, make a measurement whenever the temperature reaches a multiple of 5°C
ci	Reading 3.56 Ω off the graph, when the best-fit line hits the temperature 50.0°C,

	$\rho = A\,R\,/\,l \;=\; 1.43 \times 10^{-8} \;\times 3.56\,/\,2.50 \;=\; 2.04 \;\times 10^{-8}\,\Omega\,m$
cii	$\frac{0.01}{1.43} + \frac{0.01}{2.50} = 0.011$
ciii	(2.04 ± 0.02) Ω m
di	Systematic error, calibration error, or bias
dii	The original graph is shifted to the left compared with the corrected graph